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标题	Hybrid Pd0.1Cu0.9Co2O4 nano-flakes: a novel, efficient and reusable catalyst for the one-pot heck and Suzuki couplings with simultaneous transesterification reactions under microwave irradiation
作者	Patel A., Maity G., Pati T., Adak L., Cioffi C., Banerjee S.
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摘要（原文节选）

We report the fabrication of a novel spinel-type $Pd_{0.1}Cu_{0.9}Co_2O_4$ nano-flake material designed for Mizoroki-Heck and Suzuki coupling-cum-transesterification reactions. The $Pd_{0.1}Cu_{0.9}Co_2O_4$ material was synthesized using a simple co-precipitation method, and its crystalline phase and morphology were characterized through powder XRD, UV-Vis, FESEM, and EDX studies. This material demonstrated excellent catalytic activity in Mizoroki-Heck and Suzuki cross-coupling reactions, performed in the presence of a mild base (K_2CO …

数据表 · TABLE 5（12 行，含数值行 12 行）

Comparison of Present vs. Reported Methods for Cross-Coupling Reactions.

Sl. No.	Catalyst	Catalyst (mole%)<	Temp. (oC)	Time	Yield (%)	References
1	PdAl2O4	45	100	24 h	28 – 89	Kannan (2017)
2	Fe3O4-Pd-NHC	7.3	50	12 h	84 – 96	Stevens et al. (2005)
3	Pd/NiFe2O4	0.1	90	5 – 150 min	6 – 98	Borhade and Waghmode (2011)
4	PdCoFe2O4	3.2	Reflux	6 – 16 h	70 – 92	Senapati et al. (2012)
5	Fe3O4-DOPA-Pd	4.8	Ultra-sonication	1 – 5 min	45 – 90	Vaddula et al. (2012)
6	Pd-AcAc-Am-Fe3O4@Silica	0.28	80	1 – 3 h	80 – 98	Vibhute et al. (2020a)
7	Fe3O4@SiO2@mSiO2-Pd(II)	1.0	80	3 – 10	25 – 99.5	Le et al. (2014)
8	Fe3O4@SiO2@mSiO2 – Pd (0)	0.075	80	6 – 8	56 – 97	Li et al. (2013)
9	Fe3O4@SiO2-Pd	0.03	85	20 – 100 min	85 – 96	Khazaei et al. (2017)
10	Pd-AcAc-Am-Fe3O4@SiO2	0.3	80	4 h	62 – 96	Vibhute et al. (2020b)
11	Pd-ZnFe2O4	9.24	reflux	2 – 12 h	85 – 94	Singh et al. (2013)
12	Pd0.1Cu0.9Co2O4	4	MW/150 W	10 min	86 – 99	This work

正文对应句（逐字）：The catalytic activity of well-characterized $Pd_{0.1}Cu_{0.9}Co_2O_4$ nano-flakes was next evaluated in cross-coupling reactions. We began with the Heck coupling reaction of one-iodo-4-nitrobenzene (1) and acrylonitrile as a model system. When a mixture of one-iodo-4-nitrobenzene (1.0 mmol), acrylonitrile (1

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